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E DELL'INFORMAZIONE

Analysis of the vibration induced by the practice of running

Project report of
Data analysis for mechanical systems

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Academic year:
2023-2024

Abstract: In this study, the impact of speed and fatigue on running vibrations was investigated. Four accelerometers were placed at different levels of the subject's body to also study the transmission of vibrations through the human body. The subject had to run at two different speeds over two different periods of time. The results of the study showed mainly an increase of vibration's energy level and a subsequent reduction of vibration's transmission along the body with rising speed and fatigue' effect.

Key-words: Vibrations, running, impacts, transmissibility.

1. Introduction

During a run, while the foot is in contact with the ground, the center of mass of the runner's body is displaced downward and then upward due to the action of the leg. For the foot and the leg, the downward phase accounts for less than half of the full contact time. Indeed, they are decelerated in less than 50ms [9] whereas the contact time is about 300ms for a good runner. This leads to high decelerations that have been measured by Lafortune to up to 10g in the tibia of athletes running at 4.5m/s. [8].
These quick variations in acceleration induce vibrations of the soft tissue attached to the rigid skeletal structure [10]. It has been shown by Wakeling and Nigg that the natural frequencies of soft tissues are situated between 5 and 55 Hz which is included in the frequency range of the impacts [1, 7]. This should result in a large amplitude of vibration observed in soft tissues. However, it has been observed by Nigg and Boyer [6, 10] that the body adapts itself to control these vibrations through a concept theorized by Nigg and Wakeling called muscle-tuning [1]. According to this theory, the central nervous system can increase the damping of the vibrations by modifying the muscular activity in conditions near the resonance. Frederick and Hagy have shown that an increase of body mass leads to an increase of the amplitude of the vibrations [2].
Some external factors can also have an impact on vibrations. It has been evaluated that changing the property of the shoe insole does not modify the magnitude of the impacts [6, 10, 13], but it modifies its dynamic behavior. Indeed, Boyer highlighted that it has a strong influence on the loading rate, ie, the time derivative of the impact signal [6].

During previous studies conducted on running speed and its consequences on the human body, it has been found that there's a correlation between speed, impact shock, and the amplitude of vibrations transmitted to soft tis-

sues and to the head. It has been measured and demonstrated that the amplitude of vibrations in soft tissues (especially in muscles) increases with increasing running speed. Regardless of the muscle analyzed, it's clear that the mean power raised significantly with increasing speed. This trend is also followed by the frequencies which increase with running speed [12]. These results were obtained both for gastrocnemius medialis and for vastus lateralis and they are perfectly according to the ones achieved by previous studies [6].
Many studies focused also on another important point which is the attenuation of vibrations transmitted to the head in running. As Shorten and Winslow discussed, attenuation increases with running speeds, that's why the level of acceleration on the head is almost constant. They highlighted that the level of attenuation it's very evident for high frequencies and after 12 Hz, no significant differences in the power of bite-bar (located between the teeth) accelerations were found between different running speeds, despite large increases in the magnitude of lower extremity impact shock [13].
This result supported the hypothesis that subjects adapted to increased impact shock in a manner that reduced impact shock wave transmission.

Impact forces that result from the collision of the foot with the ground in human locomotion are repeatedly involved in many athletic activities. The application of relatively small loads over many repetitive cycles is the main cause of the high incidence of injury related to running [11].
Prolonged exposure to these repetitive loads has been associated with the development of musculoskeletal injuries such as degenerative joint diseases and athletic overuse injuries including stress fractures, some types of shin splints, cartilage breakdown, osteoarthritis, and low back pain. Regarding the lower leg muscles, the repeated impacts can cause pain and loss of function. It has also been suggested that repeated impact increases the rate of red blood cell breakdown and contributes to the depressed iron status of many distance runners [7, 13].
There is agreement that excessive impact force may produce damage to the human musculoskeletal system, however, there is a window of loading in which the biologic tissue reacts positively to the applied impact load. This short-term whole-body vibration exposure acting on cartilage, bones, ligaments, and tendons during running seems to be in an acceptable range. Moreover, has been shown that the impact forces and stresses have a positive effect on the body, including an increase in muscle force and knee extension [7, 10].

As mentioned by the literature there are several ways to analyze body vibrations during running and some of them can be more effective than others. The use of tri-axial accelerometer or motion capture system is the most widespread because it assesses respectively the variation of acceleration and displacement from an initial condition.
The advantage of skin-mounted accelerometers is to get information about three different directions, vertical, transverse and medio-lateral, while the passive filmed markers can only show the vibration displacement. Another point to take into consideration is that the positioning of accelerometers has a high influence on the results; this is given by the fact that each component of human body has its own modes of vibrations, natural frequencies and response to an impulse which can differ even between parts of the same compartment[4].
The choice of the measurement method should be taken by analysing the purpose of the investigation and consequently by identifying the bandwidth of frequencies necessary not to occur into an erroneous acquisition. Usually tri-axial accelerometer can have high sampling frequencies that can satisfy the Nyquist's theorem for running vibrations, while markers are less adaptable due to the lower sampling frequencies.
Regarding the input signal, that in case of running can be detected as the impulse generated by the contact between the foot and the ground at each step, the methods of acquisition are the following: force platform and shoe mounted accelerometer [5]. The use of the first one implies an acquisition for only one step at a time and measures the force applied by the foot on the ground, while the second can acquire a longer time history including multiple steps.

2. Methods

In this section, the methods used to evaluate the impact of running velocity on the transmission of vibrations through the body and on the left-right symmetry of the running movement are going to be explained.

2.1. Calibration Procedure

The first step of the project consisted of the calibration of Xsens Motion Trackers (MTw) which are miniature inertial measurement units containing 3D linear accelerometers, 3D rate gyroscopes, 3D magnetometers, and a barometer.

For this purpose, the shaker (TIRAvib 50101) was employed as the vibrations' source, and an additional piezoelectric accelerometer was used to compare the data coming from the Motion Tracker under investigation. To measure the actual vibration of the shaker a laser vibrometer was employed, this device allowed to acquire the speed of the vibrations which were then converted into accelerations during the processing and analysis of the data.

The MTws were fixed on the shaker by using aluminum tape and Loctite Super Attack: both the shaker and the MTw were covered with aluminum tape and then the two taped surfaces were glued together (see Figure 1). The setup was held in place until the glue completely solidified and no relative movement between the two objects was present. To obtain more valuable results four different MTw were tested at the same time, each one of the tested accelerometers was attached to the shaker following the same procedure. Also the additional piezoelectric accelerometer was placed on the shaker using aluminum tape and Loctite Super Attack.

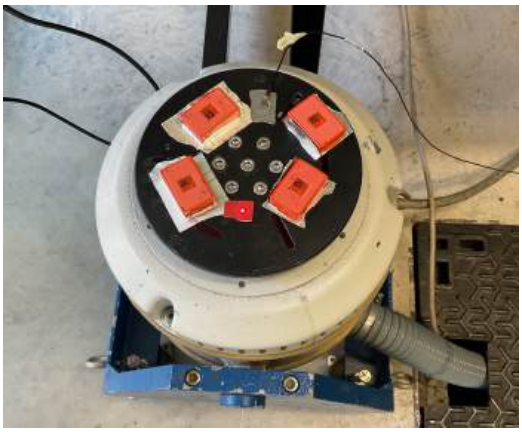


Figure 1: Setup for the calibration of XSens Motion Trackers

The shaker was set to generate a single sweep sine from 12 to 55 Hz in the vertical direction, four different levels of acceleration (1g, 2g, 5g, 7g) were tested to evaluate the response of MTws under different conditions. The frequency and acceleration tested were chosen according to the literature and according to the maximum possible displacement of the shaker employed in the calibrations. Each acceleration was tested twice to be able to perform averaging while analyzing the acquired data. The data coming from the shaker and the piezoelectric accelerometer were acquired using MeasLAB at a sampling frequency of 2048 Hz, the data coming from the MTws instead were acquired using the station provided with the motion trackers (Awinda Station) and the proprietary software (MT Manager) at a sampling frequency of 120 Hz which was the maximum sampling frequency allowed by the system.

The second part of the calibration procedure consisted of the frequency-domain analysis of the acquired data, this step was performed using Matlab. The first step of the code was to subtract the mean value from all the data coming from the MTws, that's because these IMUs, differently from piezoelectric accelerometer employed, take into account the value of the gravitational acceleration.
The second step performed in the analysis was the resampling of the data acquired from both the laser and the piezoelectric accelerometer, for this purpose the '*resample*' function of MATLAB was used. This step was necessary because of the two different sampling frequencies used during the acquisition and allowed, after the operation was done, to have all the signals with the same sampling frequency of 120 Hz.
The last fundamental step before the frequency domain analysis was to synchronize the signals coming from the two different acquisition systems used (MeasLAB and MT Manager). For this step, the cross-correlation of two signals was computed to retrieve the relative delay which was then used to shift the delayed signal.

Once the signals were all correctly resampled and synchronized, a high-pass filter with a passing frequency of 10 Hz, was applied to eliminate all the frequencies not related to the input given by the shaker. The following step was to perform the FFT using the homonym MATLAB function, after the conversion into the frequency domain it was finally possible to transform the speeds acquired with the laser into accelerations. At this point, the FRF between the shaker and all the accelerometers employed (4 MTws and the control accelerometer) was computed, firstly for each one of the two trials, and then the two obtained results were averaged. Besides the FRF, the estimators H1 and H2 were computed to compare different outcomes and validate the obtained results. For a final check and for an ulterior validation of the analysis the coherence was computed.

What was expected by the computation of the FRF and its estimators was a constant value equal to 1 along all the stimulated bandwidth (12 – 55 Hz) because the vibration generated by the shaker was the same acquired by the MTws. What was actually seen during the analysis was a decreasing value of the FRF along the considered bandwidth for each of the tested accelerations, the deviation had a maximum value of 40% (see Figure 2). This surprising result was probably caused by a change in the sensitivity which was not constant along the frequencies.

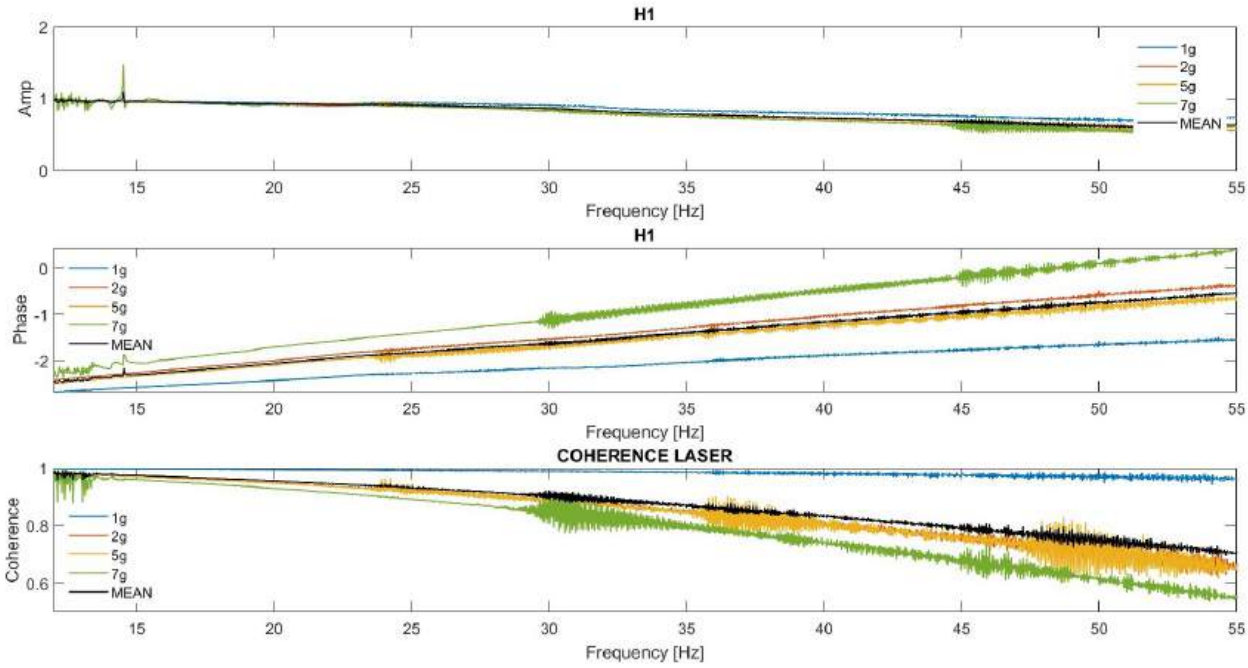


Figure 2: A decay of H1 is present for all tested accelerations

To overcome this problem, a correction curve was applied in order to get a FRF as close as possible to 1. For this purpose, the function '*polyfit*' of the 4th order was employed ($P(x) = p1 * x^4 + p2 * x^3 + p3 * x^2 + p4 * x + p5$), giving as input the difference between the obtained value and the expected values averaged on all the acceleration tested (see figure 3). The coefficients were computed with a 95% confidence bound and the values are reported in table 1. The final outcome was a unique correction function, applicable to all accelerations (see Figure 4).

Polinomial correction curve coefficients

	value	σ	$\sigma\%$
p1	$-3.98E-05$	$\pm 2.94E-06$	$\pm 7\%$
p2	0.004606	$\pm 3.95E-04$	$\pm 9\%$
p3	-0.1697	$\pm 1.89E-02$	$\pm 11\%$
p4	3.015	± 0.378	$\pm 13\%$
p5	-16.61	± 2.64	$\pm 16\%$

Table 1: Polinomial correction curve coefficients and standard deviation computed with a 95% confidence bound.

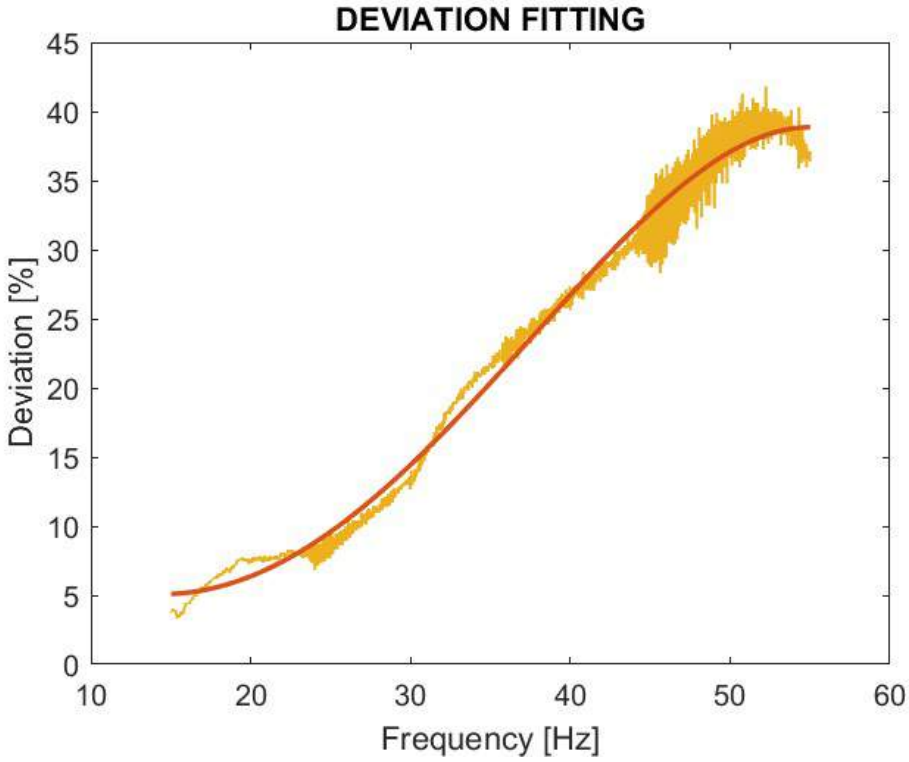


Figure 3: Fitting of the correction curve on the deviation from the expected value.

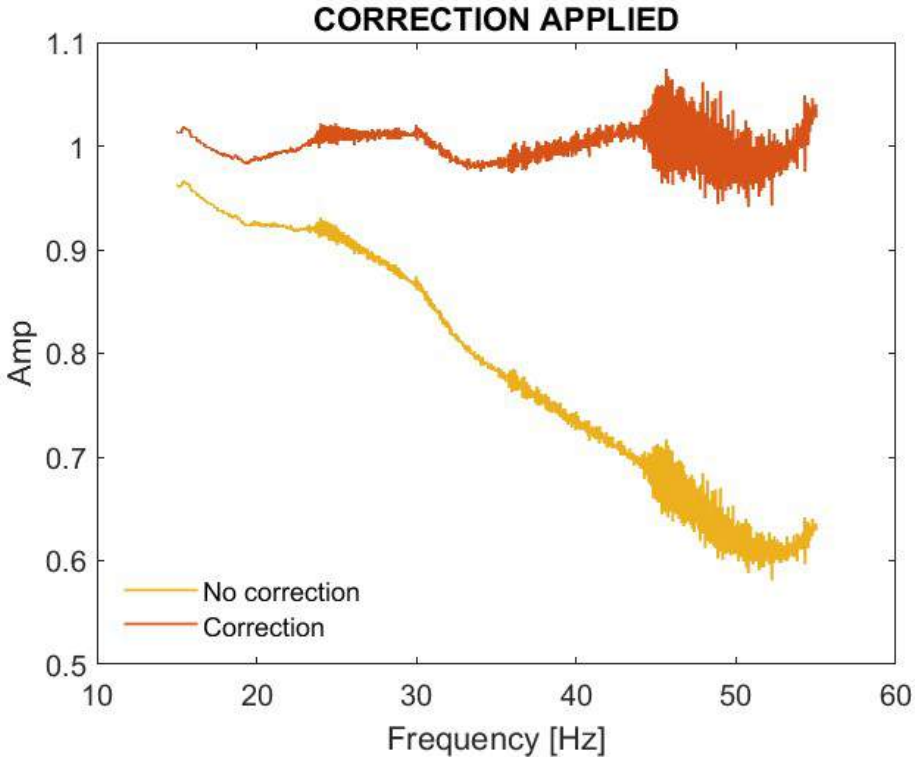


Figure 4: Correction curve applied to the averaged H1.

One last variable to be assessed was the transmissibility of the elastic bandages provided by the brand and used to place the accelerometers on the body. In this case, only one MTw was tested by attaching it to the bandage through the velcro straps present both on the back of the IMU and on one side of the elastic band; the latter was then attached to the shaker using aluminum and Loctite Super Attack as previously done . The test performed was the same as before, but using only 1g, 2g, and 5g acceleration levels. The results were analyzed using the same procedure explained above giving comparable results, it was then deduced that the bandages do

not influence the perceived vibrations.

To further assess the MTws and to check the interface between the IMUs and the shaker, the variations of the three angles (yaw, pitch and rolls) were assessed. The results showed negligible variation both during the same sweep and between different successive sweeps.

2.2. Experimental set up

For these experiments, a Technogym treadmill has been used. The choice of using a treadmill rather than doing the experiments in more "real" running conditions has been made to be able to fix constant the pace of the run. Using the treadmill there is no variability in the speed during the whole test.

Then, to evaluate the vibrations at different levels of the body, a method that is not highly invasive was needed. It appeared that using wireless accelerometers was a good compromise between the comfort of the subject and the quality of the acquisition system. Four IMUs (Xsens MTw Awinda, 16g - described in section 2.1) have been placed in different locations depending on the parameters we wanted to assess. All the accelerometers are recording data at a 120Hz frequency rate. In addition to these instrumentations, a smartwatch (Huawei Watch GT-2) was used to monitor the heart rate of the subject during the different trials.

(a) Technogym treadmill (b) Xsens acelerometer (c) Smartwatch

Figure 5: Material used for the experiments

Transmission set-up: To evaluate the transmissibility of the vibrations through the body while running the accelerometers have been placed on the right shoe of the subject, on the tibia (right below her knee), on the back (at the level of the hips) and on the forehead. These positions are highlighted with orange boxes in the figures 6 and 8.

Figure 6: Positions of the accelerometers on the subject for the transmission set-up

All three accelerometers on the body have been fixed on the subject using elastic scratch bands (see Figure 7). The one on the shoe has been fixed using aluminum tape to be able to glue the accelerometer on the surface of the shoe. For extra safety and to ensure contact of the accelerometer with the shoe, some tape has been added. This particular accelerometer will be used to evaluate the input which are the vibrations imposed to the subject through the feet.

Figure 7: A Xsens accelerometer attached to the tibia of a subject with an elastic strap

Symmetry set-up: For the second set-up, the two accelerometers have been removed from the upper body of the subject (back and forehead) and have been placed on the left tibia and the left shoe of the subject, similarly as presented in the previous paragraph.

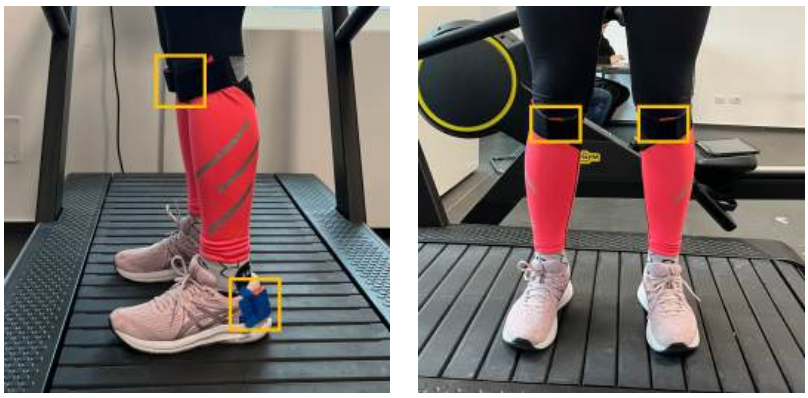


Figure 8: Positions of the accelerometers on the subject for the symmetry set-up

The reason why two different set-ups were needed is that combining these experiments would have required to have 6 or more accelerometers which would have led to a reduced sampling frequency because of the functioning of the Xsens software.

2.3. Experimental protocol

Either for the analysis of the transmission set-up or the symmetry one, the subject was asked to perform the same protocol:

1 - A one-minute walk at 3.5km/h on the treadmill. This is to set and calibrate the instrumentation.
2 - A four-minute run at 8 km/h.
3 - A four-minute run at 11 km/h.
4 - A two-minute run at 8 km/h.
5 - A two-minute run at 11 km/h.

The protocol has been first performed with the transmission set-up and then with the symmetry one.
For each speed level, two acquisitions were performed lasting 4 and 2 minutes to study the effect of fatigue, leading to four total acquisitions for each case of study.At the end of each test, the subject has been asked for a feedback about the difficulty of the run. This was done to see if there was a correlation between the perceived feelings, the fatigue effect, and the vibration level.

Note : According to the feeling of different subjects, the velocity indicated by the screen of the treadmill (3.5km/h, 8km/h and 11km/h) appeared to be faster in reality. Therefore the treadmill cannot be trusted for the speed but at least these three speed levels are different enough to have some conclusions on walk, slow run, and fast run.

2.4. Participant

In this project, only a 25-year-old female subject was tested (height = 1.64 m, weight = 63 kg). She is a healthy subject who usually runs 10 kilometers twice a week and without any particular history of injuries.

2.5. Data treatment

After the acquisition step was completed, the following step to be taken care of was the actual analysis of the data. This step, performed using MATLAB, consisted of an initial analysis of each recoded dataset on its own followed by a comparison between the different tested speeds and duration of the run considering different sub-intervals.

The first step required in the analysis was the conversion the accelerations acquired along the three axes, referred to the reference system of each motion tracker, to the global reference system. The latter was specified on the Xsens data sheet along with the conversion matrix, reportet in Figure 9, which allows passing from local to global reference system by using the angles (yaw, pitch, and roll) acquired by each motion tracker during the tests. The global reference system was the same for each accelerometer and was defined with the x-axes pointing to magnetic North, the y-axes pointing to West and the z-axes perpendicular to the ground.

$$
\begin{aligned}
R_{GS} &= R_{\psi}^{Z} R_{\theta}^{Y} R_{\phi}^{X} \\
&= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}
\end{aligned}
$$

Figure 9: Rotation matrix to pass from local to global reference system as found in the Xsens data sheet. In the picture Ψ is the yaw angle, Θ is the pitch, and Φ is the roll.

Following this step, a filtering of all the signals was performed by applying a band-pass filter with a bandwidth between 5 and 55 Hz to extract only the frequency related to the act of running; the bandwidth was chosen according to what was found in the literature.
From this point on, the analysis was carried on considering 1-minute sub-histories for each acquisition and for each direction (x-, y-, and z-axis) in order to obtain multiple sub-sets of data which could then be compared. To determine each time window the starting and ending points were manually selected taking care that they were not chosen during a vibration peaks. Subsequently, for each time window, the peaks were detected using MATLAB's function '*findpeaks*' and setting for each axis and for each sub-set a proper threshold value. Each peak was then used to divide the signal (from now on the selected time windows are referred to as the signal) into sub-sets which allowed, later on, to perform averaging of the analyzed data. It was chosen to consider 10 samples before and 14 samples after the identified peak in order to consider all the vibrations related to each input without considering the ones caused by the symmetrical feet. The motion trackers placed on the back and on the head, in fact, not only were subjected to the inputs given by the right foot but also to the ones coming from the left one. A direct consequence of using such short sub-sets ($25 samples \times 1/120 Hz = 0.2083 s$) was the resolution in the frequency domain ($1/0.2083 s = 4.8008 Hz$). For this reason, the sub-sets were elongated adding 200 zeros before and after each actual sub-set obtaining a resolution in the frequency domain of 0.2824 Hz.

Once the sub-sets were all created the following step was to finally perform the FFTs, in order to obtain the spectra and all the related parameters. The FFTs were performed before for every single sub-set and then averaged between all the different sub-histories. Other than the FFT the auto spectra end the cross-spectra between the input and each of the other outputs were computed.
Once the spectra were computed and transformed to the single-sided version, the correction curve found during the calibration process was applied to take care of the decay in the sensibility of the motion trackers. The final step for this first part of the analysis consisted in computing the FRF and its estimators between the input and each of the accelerometers used and the coherence. This last step was only computed for accelerations on the z-axis because, as found in the literature, they are the most affected by the impact caused by running.

After computing the single spectra FFTs, the weighting curves found in the ISO 2631 and referred to the whole body vibrations were applied to the spectra of each direction, as specified in the norm. The choice of using the curve referred to as whole-body vibration (WBV), instead of considering feet-transmitted vibrations (FTV) it's due to the fact that there is no ISO standard available for the latters. Once the curve reported in Figure 10 was applied, the Power Spectra and Power Spectral Density for each direction were computed. Since the addition of zeros to expand the time window reduces the power of a factor $\frac{N}{N+L}$, where N is the number of actual samples comprised in each peak (10 before + peak + 14 after the peak = 25 actual samples) and L is the number of added zeros (200 before + 200 after = 400), the inverse of this factor was applied to the power spectra computation to obtain true powers of the nonzero data. The same approach here reported was also followed by Shorten et. al. [13] during their analysis which, for some parts, resembles the one here reported.
One last step consisted of computing the global parameters along all the directions at the same time by summing the squared values and then extracting the square root. Finally, the RMS was computed and the obtained values were compared to the standard using the A(8) coefficient. In order to compute this coefficient, the effective vibration time has been considered, excluding the period when the vibration level was negligible. In the last part, the total duration time that the subject could run without risks was computed considering the limit A(8) value of the ISO standard. This was done both for the slow and the fast pace.

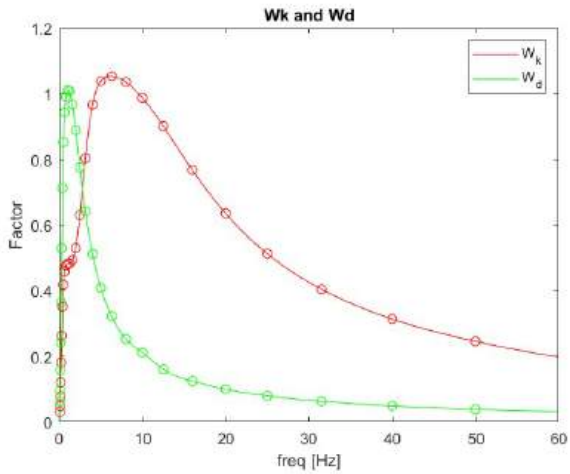


Figure 10: Weighting curves. Wk for z-axis, Wd for x and y axis

3. Results and discussion

3.1. FFTs

The first results obtained with the MATLAB code were the FFTs of all the four signals acquired (input,leg,back and head) only in the vertical direction. The FFTs computed for each step were averaged in a time frame of one minute by either using the RMS averaging or complex averaging in order to see possible inconsistency between the two methods. Afterwards all the FFTs were corrected with the calibration curve of the accelerometers. The results showed a continuity along all the cases analyzed for the speed and fatigue effect, with higher values of amplitudes in the range of frequencies between 5-20Hz as expected by the literature [3, 5]. The only issue related to this part was the FFT curve of the accelerometer placed on the shoe: the amplitudes of the bandwidth previously mentioned were expected to be higher in the input than in all the other signals, while they resulted to be lower or comparable. Moreover a complete drop of amplitude around the 15Hz and 45Hz frequencies wasn't expected. The possible cause could have been the damping effect of the shoe itself that was not mentioned in any of the articles in which a heel-cup mounting of the accelerometer was adopted. In the following results of this study the issue that has just been proposed will be taken into account.

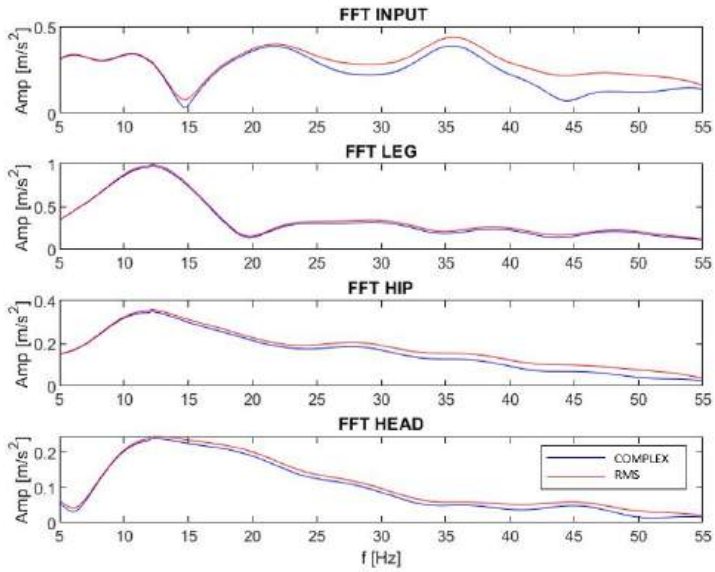
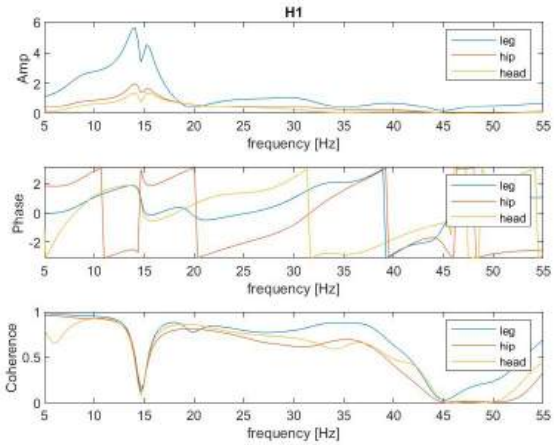


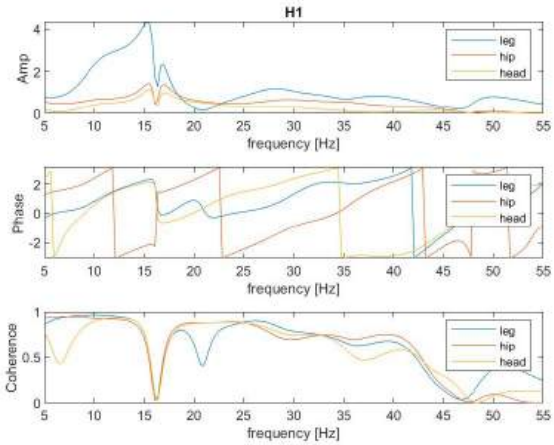
Figure 11: Example of FFT curves for all the signals averaged with RMS and complex methodology

3.2. FRFs

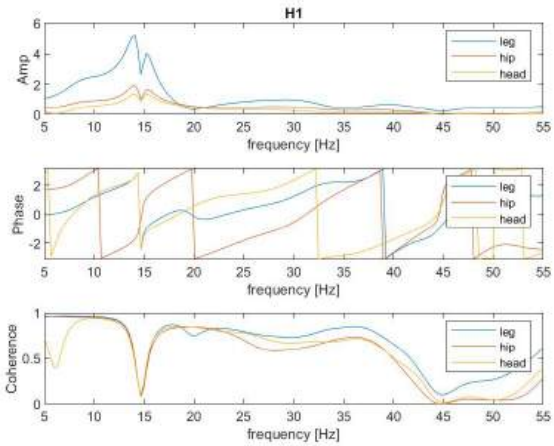
The next step of the study was the computation of the frequency response function between the input and the other acquisition's points in order to analyze the transmission of the vibrations through all the body in the cases of changing speed and fatigue's effect. The graphs of H1 estimator's amplitude, phase and coherence have all been considered to see the reliability of the data. In all the following figures is possible to see a drop of coherence in the frequencies cited before (15Hz and 45Hz) which are related to the acquisition of the input' signal.
As first result it has been proved that from the leg up to the head the level of vibrations reduces due to the body acting as a damper: this can be seen in the amplitude's graphs of H1 in which the curve referred to the head is always below the one of the hip and leg in the bandwidth of 5-20Hz. In the case of speed's effect it has been found that always in the bandwidth of interest (5-20Hz) the amplitudes decrease and the peak frequency increases at faster speeds.
Instead for the fatigue's effect it has been shown that between 5-20Hz the values of amplitude decreased from the first minute to the last one of running for both the cases while the peak frequency slightly moved to the right. This last result was found for the fast pace, while for the slow one, no differences can be observed.
The previous considerations are in agreement with what read in the literature, in which the cause is attributed to the tuning of muscles with increasing speed and time of running.



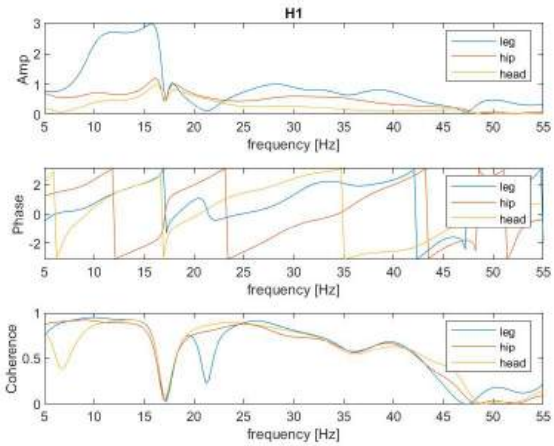
(a) H1 slow pace (first min)



(b) H1 fast pace (first min)



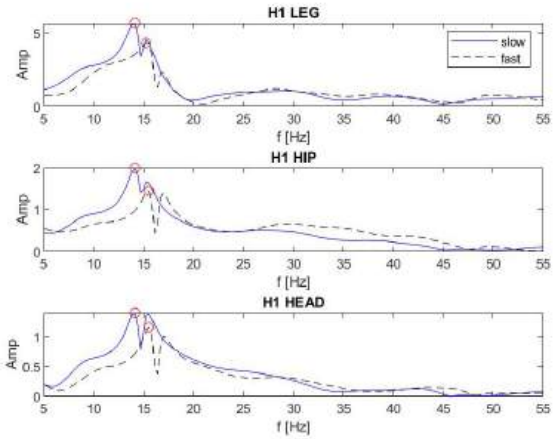
(c) H1 slow pace (last min)



(d) H1 fast pace (last min)

Figure 12: H1 of leg, hip and head at different speeds taken at first and last minute of running

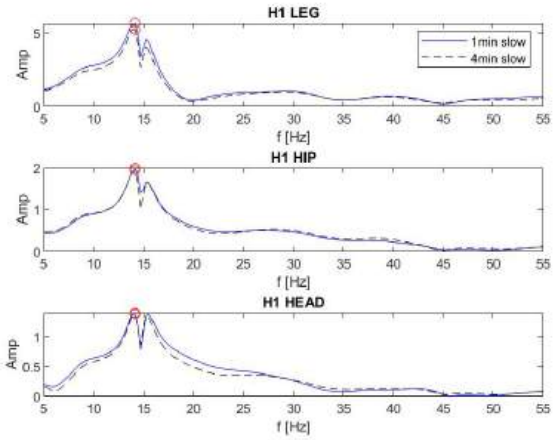
In order to better show the previous results the H1 peak value of each case has been analyzed to display the percentage variation of amplitude (Delta) and the peak frequency shift for both speed and fatigue's effect.



(a) Speed's effect

Body Part	p SLOW [EU]	p FAST [EU]	Delta [%]	f SLOW [Hz]	f FAST [Hz]
LEG	5.6505	4.3157	23.6223	14.1176	15.2471
HIP	1.9987	1.4522	27.3414	14.1176	15.5294
HEAD	1.4040	1.1635	17.1244	14.1176	15.5294

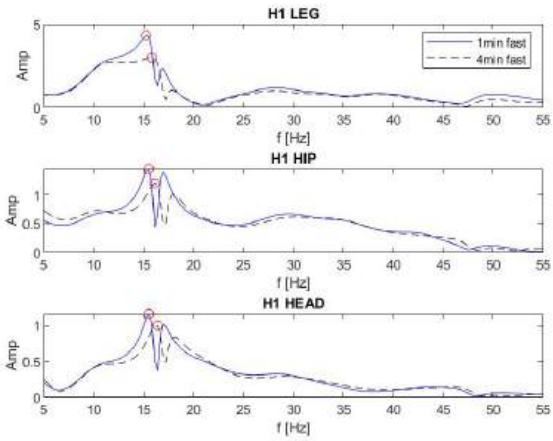
(b) Speed's effect table



(c) Fatigue's effect at slow speed

Body Part	p 1min [EU]	p 4min [EU]	Delta [%]	f 1min [Hz]	f 4min [Hz]
LEG	5.6505	5.2052	7.8797	14.1176	14.1176
HIP	1.9987	1.9491	2.4828	14.1176	14.1176
HEAD	1.4040	1.3821	1.5544	14.1176	14.1176

(d) Fatigue's effect table



(e) Fatigue's effect at fast speed

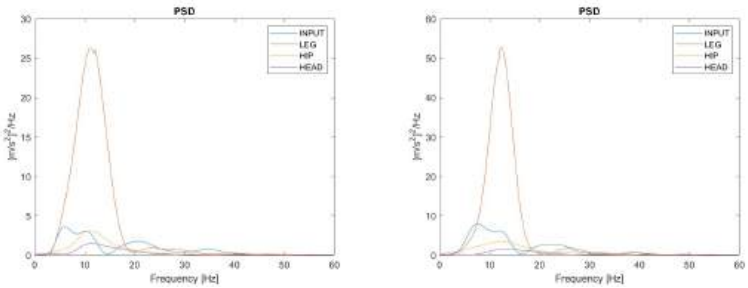
Body Part	p 1min [EU]	p 4min [EU]	Delta [%]	f 1min [Hz]	f 4min [Hz]
LEG	4.3157	2.9841	30.8547	15.2471	15.8118
HIP	1.4522	1.2069	16.8957	15.5294	16.0941
HEAD	1.1635	0.9997	14.0772	15.5294	16.3765

(f) Fatigue's effect table

Figure 13: Analysis of speed and fatigue' effect on H1 amplitudes and frequencies

3.3. PSD

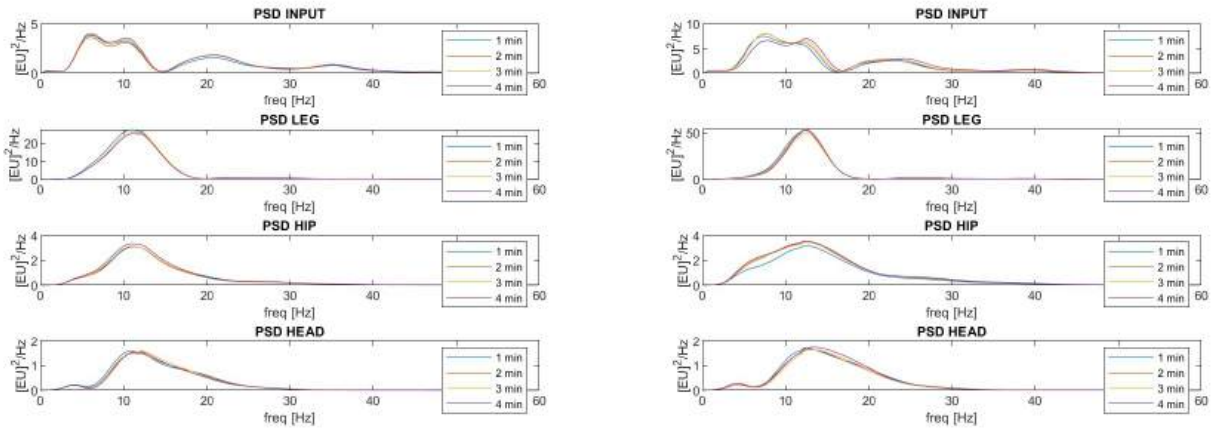
After the computation of the global autospectra (coming from the square root of the sum of autospectra from all the 3 axis, each one at the second power), the PSD has been evaluated as described in the previous section. Figure 14 shows the Power Spectrum Density (PSD) taken from the middle minute of the 4-minute run for both the speeds.



(a) PSD slow pace (b) PSD fast pace

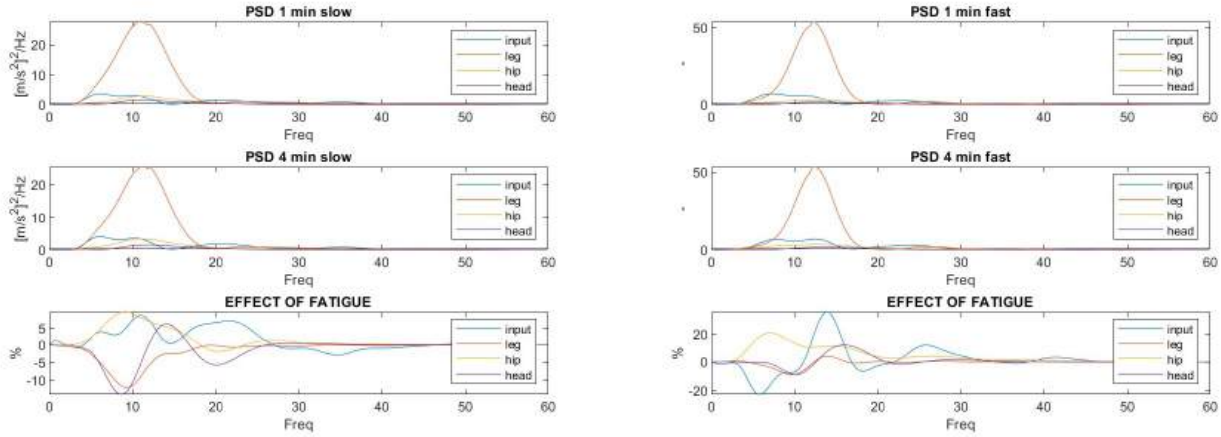
Figure 14: Differences betweeen the PSD of the different accelerometers at different speed

Figure 15 displays the PSD of the 4 different minutes of the run overlapped in order to highlight the differences due to fatigue. Increasing the time, a slight increase in amplitude can be seen. Moreover, the frequency of the peaks tends to shift to higher frequencies. To better compare the results obtained, the relative increment of the PSD amplitude has been computed and it's reported in Figure 16. After the 10 Hz threshold, the PSD has higher amplitudes in the last minute compared to the first one. This difference has a maximum of about 25% considering the fast pace. For the slow pace, the trend is similar but the increment is lower.



(a) PSD slow pace (b) PSD fast pace

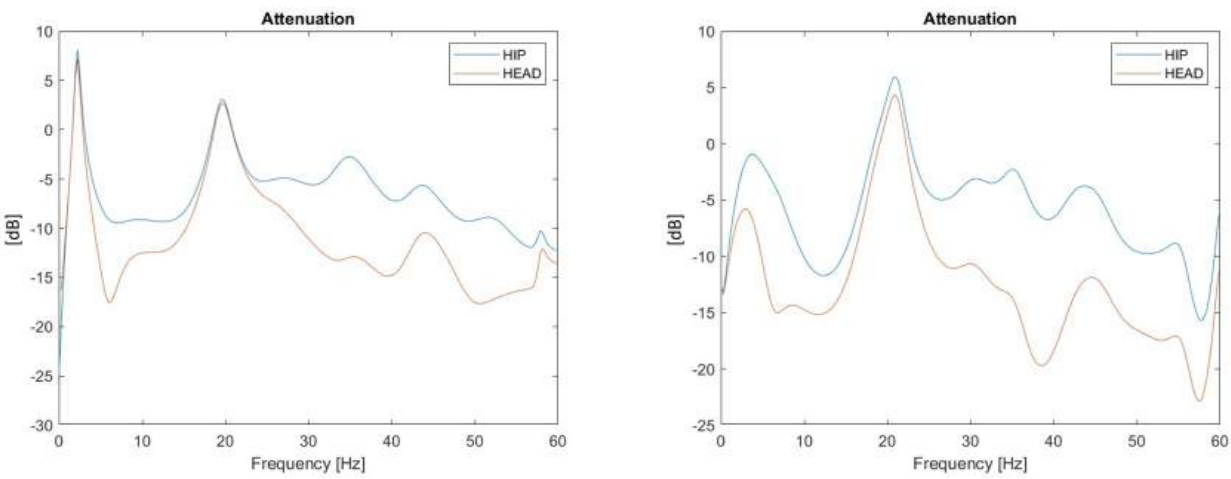
Figure 15: Comparison of the PSD considering different minutes



(a) Slow pace (b) Fast pace

Figure 16: Effect on fatigue for both the paces

The PSD was used also to compare the energy content between the leg and the accelerometers placed on the hip and the head. In this case, the input was not considered because of the damping effect of the shoe. The attenuation was computed using a dB scale and here the results obtained for the middle minute of the run for both the paces are reported in Figure 17. Since there's a drop of the PSD of the leg in the range of 20 Hz and in that region, both the PSD of the head and the one of the input are higher than the leg one, here we can see a huge peak. Considering the rest of the frequency range, it's clear that there is a higher attenuation on the head compared to the hip. These results were almost equal for each minute of a certain pace, so it seems that the attenuation between the leg and other locations has no dependence on the time.



(a) Slow pace (b) Fast pace

Figure 17: Attenuation for both the paces

3.4. RMS and A(8)

The last evaluation was done computing the A(8) coefficient. It was computed starting from the RMS value of each accelerometer, multiplying it by the square root of the ratio between the effective time of the run and the whole duration of a typical working day (8 hours). The ISO 2631 suggest that the limit for comfort is 2 $m/(s^2)$. The results obtained for this study are shown in Figure 18. As it's possible to see, the highest values of A(8) are reached in the leg and the fast speed has higher values compared to the slow one except for the head. As final result, it was found that the limit for comfort of the subject is exceeded after 46 minutes of running taking in consideration the worst case, i.e the energy level of vibrations on the leg at fast pace.

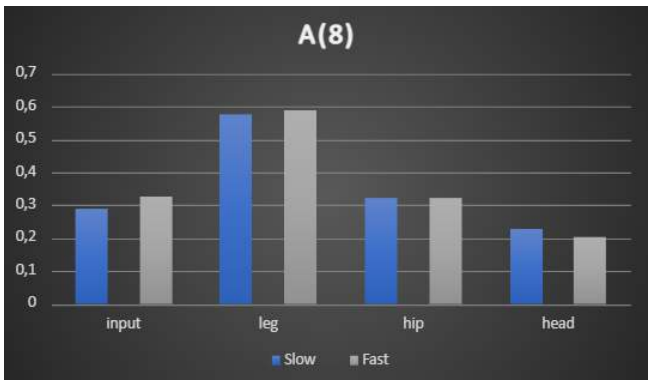


Figure 18: A(8) for Input, Leg, Hip and Head

4. Conclusions

4.1. Effect of running speed

On the FRF: As first result of this study is possible to conclude that the speed of running is an important factor for vibration's transmission through human body. As shown by the FRF analysis, an increasing speed can lead to a shift of the peak frequency towards higher frequencies, but first of all to a decrease of the vibration's amplitudes transmitted from the leg up to the head. It can also be seen by the tables of Figure 13 that the attenuation is more prominent in the leg and back, while in the head the level of vibrations tends to be reduced less or kept constant. As mentioned in previous paragraphs and in the literature, the effect of reduction of transmission is a response of the human body to increasing vibrations. The body tries to limit their effects by tuning the muscles and enhancing the damping response that leads to have lower amplitudes.

On the PSD and attenuation: To understand the impact of speed on the PSD of the different signals, it is interesting to compare the frequency and the amplitude of the highest peak in the PSD presented in Figure 14. In the table 2 are reported the frequency shift and the amplitude gain in % induced by the speed change.

	Input	**Leg**	Hip	Head
Frequency shift	17%	9%	10%	15%
Amplitude gain	56%	50%	10%	-8%

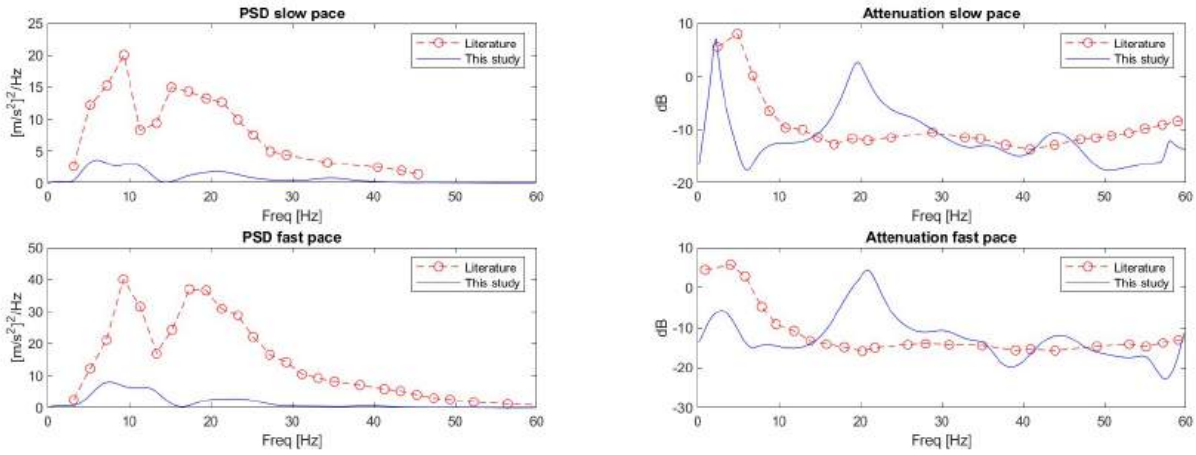
Table 2: Impact of the speed on the frequency and amplitude of the signal for each accelerometer position

The general trend is that increasing the speed of the run will result in a shift of the frequency peak to the right, ie an increased frequency of the peak of about 10% to 17%.

The other result that can be highlighted from this table is the impact on the amplitude of the PSD. Indeed, increasing the speed results, most of the time in the presented study in a gain of amplitude. This gain tends to reduce as the vibrations are registered far from the impact point. As a matter of fact, the amplitude of the first peak of PSD increased by 56% with the increased speed for the input accelerometer, this gain is reduced to only 10% for the hip one, and for the head the tendency is even inverted. The amplitude of the first peak was higher at the slow pace for this particular accelerometer.

This amplitude and frequency gain while increasing the speed that has been highlighted is in accord with the published literature that observed the same trend [6, 12, 13]. Moreover, Shorten and Winslow have also suggested in their paper that the more fast the run is, the more the head's vibrations are damped. This should result in a constant level of vibration in the head [13]. In our case, for the two different paces, the Head's vibration PSD maximum amplitude was 1.46 $(m/s^2)^2/Hz$ for the slow pace and 1.35 $(m/s^2)^2/Hz$ for the fast one.

Here are reported in the same graphs the results from Shorten and Winslow and these of the present study :



(a) Input PSD Comparison

(b) Head attenuation Comparison

Figure 19: Comparison with the literature [13]

As explained in the previous paragraphs, the trends are the same. Observations that can be made is that in this study, the PSD amplitudes are lower than the one found in the literature, this can be a result of the position of the input accelerometer. Indeed, as explained, in section 2.2, the input accelerometer has been placed on the subject's shoe and not on its body. The amplitude drop can be a result of the damping effect of the shoe. To make a more relevant comparison, the two attenuations that are compared in figure 19b are leg-head attenuation from this study and ankle-head attenuation from Shorten and Winslow's study. If the particular shape cannot be analyzed because of the different set-ups, it can be noticed that the order of magnitude are the same.

4.2. Effect of fatigue

To evaluate the effect of fatigue it was decided to also take into consideration the physiological parameter of the heart rate (HR) to try to assess to actual physiological fatigue. In Figure 20 are reported the data acquired by the smartwatch during the 4-minutes run.

(a) HR of 4-minute run at slow speed

(b) HR of 4-minute run at fast speed

Figure 20: Heart rate at different speeds for the 4-minute runs.

Considering the average HR value provided by the smartwatch used to evaluate this parameter, and reported in table 3, it is possible to see how those values are quite similar for the slow and fast pace (146 bpm for the first and 151 bpm for the second) differing only by 3% from each other.

	Min HR	**Average HR**	**Max HR**
4-minute slow	81	146	159
4-minute fast	93	151	168

Table 3: Values of heart rate provided by the smartwatch during the different runs.

In both cases, the mean value is within 70-79% of the HRmax, estimated for the subject as $220 - age = 220 - 25 = 195bpm$. In this HR band, physical activity can be sustained for 10 to 40 minutes (see Figure 21), so, even the 4 minutes of fast-speed running cannot properly be considered as a state of fatigue. Anyway, is

possible to make some consideration by comparing the acquisition related to the first and the last minutes of the same 4-minute run; for the purposes of evaluating fatigue, the 2-minute run won't be evaluated.

ZONA	5	Massima Intensità	**90% - 100% della FCMax**: altissima intensità. Migliora la prestazione massima e la velocità. Migliora la velocità massima di sprint. **CONSIGLIATO PER**: resistenza anaerobica e potenza muscolare.	0 - 2 minuti
ZONA	4	Zona Anaerobica	**80% - 89% della FCMax**: migliora la resistenza alle alte velocità e la prestazione massima. Consente di innalzare la propria soglia anaerobica. **CONSIGLIATO PER**: allenamento potenza e miglioramento massima prestazione.	2 - 10 minuti
ZONA	3	Zona Aerobica	**70% - 79% della FCMax**: migliora la condizione aerobica e ritarda la fatica causata dall'acido lattico. Espande i vasi sanguigni, aumenta la capacità polmonare e rinforza il muscolo cardiaco permettendo allenamenti più lunghi. **CONSIGLIATO PER**: resistenza	10 - 40 minuti
ZONA	2	Controllo del Peso	**60% - 69% della FCMax**: aumenta l'efficienza con cui grassi e carboidrati sono usati come fonti energetiche. Prepara il corpo ad allenamenti più intensi. **CONSIGLIATO PER**: dimagrimento.	40 - 90 minuti
ZONA	1	Intensità Moderata	**50% - 59% della FCMax**: migliora la circolazione sanguigna, aiutando ad avere un cuore sano e a recuperare dopo allenamenti intensi. **CONSIGLIATO PER**: riscaldamento e defaticamento.	A lungo
ZONA	0	Riposo	**meno del 50% della FCMax**: riposo	Molto a lungo

Figure 21: Training zones used to assess the different exercise intensity range via HR.

From the comparison between the first and the last minute of H1 (reported in Figure 13) considering both speeds is possible to highlight how from the first to the last minute in the fast-speed run there is a noticeable shift in the frequency of the peak towards higher frequency (see Figure 13f). This same behavior is not reported in the slow-pace running, probably due to the short duration of the run.
An even more noticeable difference between the graphs displayed in Figure 13 is the change in the amplitude of the graphs. The variation between the first and the last minute is once again more present in the fast-speed running where, with the increasing of time there is a decrease of the amplitudes of the peaks for each of the locations considered (the highest decrease is shown for the H1 between the input and the leg with a value of 30.85%, for other values see Figure 13f).

These variations between the first and the last minute can be explained by watching at the graphs reported in Figure 16 where is possible to see how the PSD increases from the first to the last minute. This result is in accordance with what was found in the literature, at these purposes the article written by Friesenbichler et al. [3] can be mentioned. In their experiment, subjects ran as far as possible maintaining a constant pre-established running pace, 50 steps were taken from the first and the last part of the experiment to evaluate changes in the vibrations as fatigue arose.
In the article, the fatigued state was associated with higher energy content if compared to the first part of the running, for the present study, the increased energy content is portrayed by the positive percentage variations displayed in figure 16. The increase in vibrations, and so in energy content is because muscle tuning, responsible for the damping of the vibration through muscle contraction, is highly affected by fatigue: as the running is carried on for a longer time interval the muscular fatigue increases reducing the ability of muscle to contract in response to vibrations.

By comparing the slow- and fast-speed PSD (see fig. 16) is possible to see how the increase was very similar for each location in the first, while at higher speeds the input underwent a bigger increase than leg, back, and head. This behavior can explain the differences between the H1 for the fast and the slow pace. At the lower speed there is no significant difference between locations, so the ratio between them is held almost constant showing no time dependence, H1 for the first and last minute is almost the same (see Figure 13c). When considering the fast speed is possible to see how, even if all the PSDs show a positive deviation, the input has a way bigger increase (25% of the input compared to 15-20% of other locations) which affects H1 with a decrease as the time interval increases (input is at the denominator).
Another factor that can help explain the different behavior of fatigue at the two evaluated speeds is, once again, the HR. By looking at the graphs reported in figure 20 is possible to see how for the slow speed the HR reaches an almost constant value, while the HR acquired during the fast-speed run has a slow and constant increase. Also in this second case, the maximum HR value is 11% bigger than the mean value, while for the slow pace, this increase is only 8%. One last comment on the difference between the fatigue effect at the two tested speeds could be made considering how the subject rated the level of physical activity in the two cases. For the slow pace she gave a value of 5 out of 10 do describe the effort and the difficulty of the task, while for the fast speed, the value was of 8 out of 10. This different percieved fatigue was also found, as illustrated above, in the results.

In conclusion, even if the experimental setup could not define a proper fatigue state some consideration, in accordance with what was found in the literature, can be made regarding fatigue effect.

4.3. Limits

The first limitation to be evaluated in the present study is the absence of comparison between different subjects as only one participant took part in the research. To take this limitation into account, comparisons were made considering the literature on the topic, mainly [13].
Another point that requires some clarification, and which has already been partially explained at the time, is the difference between the speed perceived from the subject and the one set on the treadmill. Since the actual value of the speed was not so relevant in the present study, which was more focused on the comparison between two speeds defined as fast and slow by the participant, this issue was not studied in detail. Regarding the set-up, another possible source of error could be the attachment of the motion tracker to the shoe. By comparing the FFTs of the signal acquired at the shoe and at the knee level, is possible to notice how some frequencies are present in the latter, but are attenuated in the former. This could be caused by the attenuation of vibrations provided by the shoe since only in this case there is an attenuation and this was the only accelerometer not mounted directly on the subject's body. A clarification is needed regarding the positioning of this accelerometer; the placement, in this case, was chosen following what was found in the literature [3, 5] in the attempt to emulate other experimental setups to obtain comparable data.
Another limitation encountered during the experimental procedure was related to the sampling frequency of the motion trackers (120 Hz), which was rather low and allow only frequencies below 60 Hz to be correctly analyze. The problem related to sampling frequency arose because the range of frequencies involved during running were found in the literature to be in the range of 5-55 Hz. Still related to the sampling frequency is the problem explained in the data analysis; low sampling frequency means fewer and sparser samples for each seconds which limited the time frame of each subset increasing, as a consequence, the resolution in the frequency domain. In addition the maximum sampling frequency allowed by the motion trackers depends on the number of tracker used at the same time and decreases as the number of employed trackers increases. In the case of the presented study, this limited the amount of IMUs that could utilized during one acquisition and so the study of the damping effect of the body (only few locations could be selected).
As a final limitation is necessary to report that there is no current standard for the study of the vibrations transmitted through the feet nor the vibration occurring during running. In the present study the ISO 2631 was used as a reference considering the case of whole body vibrations. Still related to the norm, in this case the A(8) index was used to assess the danger of the vibrations for the runner but its reliability should be taken with a grain, because running can not be considered as an ordinary job activity. Further studies could expands the knowledge of the effects of vibration occurring during running and, more in general, related to sports activities.

4.4. Future developments

As the reader has probably already noticed, the experimental setup section describes two different setups: transmission and symmetry, while only the former has been evaluated in the data analysis chapter. For this reason, a future development of this study could focus on the symmetry between the right and the left leg comparing how the symmetry, if present, is affected by fatigue and by changes in speed. Another point that could be more precisely and accurately investigated is the fatigue effect, longer runs would be necessary and also other parameters, in addition to heart rate should also be considered to correctly evaluate the onset of fatigue. Such other parameters could be, for example, the lactate level and/or the percentage of VO2max.

References

[1] J M Wakeling B M Nigg. Impact forces and muscle tuning: a new paradigm. *Exercise and Sport Sciences Reviews*, 29:37–41, 2001.

[2] J. Hagy E. Frederick. Factors affecting peak vertical ground reaction forces in running. *International journal of sport biomechanics*, 1986.

[3] Bernd Friesenbichler, Lisa M. Stirling, Peter Federolf, and Benno M. Nigg. Tissue vibration in prolonged running. *Journal of Biomechanics*, 44(1):116–120, 2011.

[4] Benno M. Nigg Katherine A. Boyer. Soft tissue vibrations within one soft tissue compartment. *Journal of Biomechanics*, 39:645–651, 2006.

[5] Benno M. Nigg Katherine A. Boyer. Quantification of the input signal for soft tissue vibration during running. *Journal of Biomechanics*, 40:1877–1880, 2007.

[6] Benno M.Nigg Katherine A.Boyer. Muscle activity in the leg is tuned in response to impact force characteristics. *Journal of Biomechanics*, 37:1583–1588, 2004.

[7] Arash Khassetarash, Reza Hassannejad, Mir Mohammad Ettefagh, and Vahid Sari-Sarraf. Fatigue and soft tissue vibration during prolonged running. *Human movement science*, 44:157–167, 2015.

[8] Mario A. Lafortune. Three-dimensional acceleration of the tibia during walking and running. *Journal of Biomechanics*, 24:877–879, 881–886, 1991.

[9] B M Nigg M F Bobbert, H C Schamhardt. Calculation of vertical ground reaction force estimates during running from positional data. *Journal of Biomechanics*, 24:1095–1105, 1991.

[10] Benno M Nigg. Impact forces in running. *Current Opinion in Orthopaedics*, 8(6):43–47, 1997.

[11] Tom F Novacheck. The biomechanics of running. *Gait & posture*, 7(1):77–95, 1998.

[12] C.Hautier R.Trama, Y.Blache. Effect of rocker shoes and running speed on lower limb mechanics and soft tissue vibrations. *Journal of Biomechanics*, 82:171–177, 2019.

[13] Martyn R Shorten and Darcy S Winslow. Spectral analysis of impact shock during running. *Journal of Applied Biomechanics*, 8(4):288–304, 1992.